

Basic Counting

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COMP2711 Discrete Mathematical Tools

Counting can determine how many outcomes are possible or what repetition is unavoidable. For an exact count, specify one outcome and count every valid outcome exactly once. For a guarantee, identify the objects and the boxes.

1 Basic counting rules

1.1 Product rule

MUST UNDERSTAND Multiply the numbers of choices in successive stages.

If stage i has n_i choices for every valid sequence of earlier choices, the number of complete outcomes is

$$n_1 n_2 \cdots n_k.$$

Example: Student IDs. An ID has two uppercase letters followed by three digits; the first digit cannot be zero. Repetition is allowed. The number of IDs is

$$26 \cdot 26 \cdot 9 \cdot 10 \cdot 10 = 608,400.$$

Example: Functions. Let $|U| = m$ and $|V| = n$. Each input has n possible outputs, so there are n^m functions $U \rightarrow V$. For an injection, outputs cannot be reused:

$$\#\{\text{injections } U \rightarrow V\} = \begin{cases} n(n-1) \cdots (n-m+1), & m \leq n, \\ 0, & m > n. \end{cases}$$

For $U = \{a, b\}$ and $V = \{1, 2, 3\}$, this gives $3^2 = 9$ functions and $3 \cdot 2 = 6$ injections. Empty products equal 1; there is one function from an empty domain, including when the codomain is empty.

1.2 Sum rule

MUST UNDERSTAND Add the sizes of disjoint cases.

If the outcomes split into disjoint cases of sizes $n_1, \dots, n_k$, then the total is

$$n_1 + \cdots + n_k.$$

Example: Getting to campus. Choose either a bus or a ferry. There are three bus routes, each with four departure times, and two ferry routes, each with three departure times. The number of route–time plans is

$$\underbrace{3 \cdot 4}_{\text{bus}} + \underbrace{2 \cdot 3}_{\text{ferry}} = 18.$$

1.3 Complement counting

MUST UNDERSTAND Subtract the forbidden outcomes from all outcomes.

For $B \subseteq A$,

$$|A \setminus B| = |A| - |B|.$$

For two possibly overlapping forbidden sets $B, C \subseteq A$, subtract both and add back their intersection:

$$|A \setminus (B \cup C)| = |A| - |B| - |C| + |B \cap C|.$$

This is the two-set inclusion–exclusion principle.

Example: Passwords. A password has six characters, each an uppercase letter or digit. Passwords with at least one digit are all passwords minus all-letter passwords:

$$36^6 - 26^6 = 1,867,866,560.$$

Choosing a digit position first would count passwords with several digits more than once.

1.4 Division rule

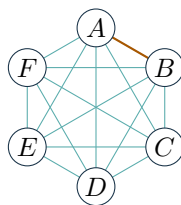
MUST UNDERSTAND Divide only when every outcome has the same number of descriptions.

If there are N descriptions and each outcome has exactly d descriptions, then the number of outcomes is N/d .

Example: Six-person handshakes. Every pair of distinct people shakes hands once. Choosing the first person and then the second gives $6 \cdot 5$ ordered pairs. The descriptions (A, B) and (B, A) represent the same handshake, so the count is

$$\frac{6 \cdot 5}{2} = 15.$$

In the complete graph K_6 below, vertices represent people and edges represent handshakes.

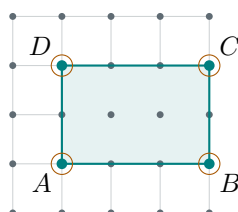


Example: Rectangles on a grid. An $n \times n$ grid has n rows and n columns of equally spaced points, where $n \geq 2$. Count rectangles with grid-point vertices and horizontal or vertical sides; squares are included.

Choose a starting vertex in n^2 ways, a different column in $n - 1$ ways, and a different row in $n - 1$ ways. These choices determine the rectangle. Each rectangle has four possible starting vertices, so

$$\frac{n^2(n-1)^2}{4}.$$

For $n = 5$, there are 100 rectangles. Here n counts points along a side; an $n \times n$ board of square cells has $n + 1$ points along a side.



Exercise: Squares on a grid. On the same $n \times n$ grid of points, how many squares have grid-point vertices and horizontal or vertical sides? Left as an exercise.

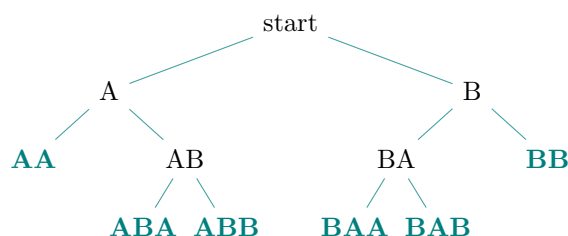
1.5 Tree diagrams

A root-to-leaf path describes one complete outcome when later choices depend on earlier ones.

Example: Best-of-three series. Teams A and B play until one wins two games. There are six complete sequences:

$AA, \quad ABA, \quad ABB, \quad BAA, \quad BAB, \quad BB.$

There is no sequence AAA : play stops after AA .



2 Pigeonhole principles

MUST UNDERSTAND More objects than boxes force a collision.

If $k + 1$ objects are assigned to k boxes, some box contains at least two. Otherwise each box would contain at most one object, allowing at most k objects in total.

For example, among 27 English words, two have the same initial letter: 27 words are assigned to 26 letters. Similarly, for finite sets U, V with $|U| > |V|$, no function $U \rightarrow V$ is injective.

2.1 Generalized pigeonhole principle

If $n \geq 0$ objects are assigned to $k \geq 1$ boxes, then

some box contains at least $\lceil n/k \rceil$ objects.

Proof. If every box held at most $\lceil n/k \rceil - 1$ objects, the total would be at most

$$k(\lceil n/k \rceil - 1) < n,$$

a contradiction.

To force at least $r \geq 1$ objects into some box, the sharp threshold is

$$k(r - 1) + 1.$$

With $k(r - 1)$ objects, each box could hold exactly $r - 1$; one more forces r .

The distribution need not be balanced. For ten objects and four boxes, $\lceil 10/4 \rceil = 3$. Both $(3, 3, 2, 2)$ and $(7, 1, 1, 1)$ are possible. Only one box reaches three in the second arrangement. The theorem guarantees at least one such box, not $n \bmod k$ boxes.

Example: Some suit versus hearts. Draw without replacement from a standard 52-card deck. To guarantee three cards of some suit, eight cards are insufficient (two per suit), but

$$4(3 - 1) + 1 = 9$$

suffice. To guarantee three hearts specifically, all 39 non-hearts could appear first, so the threshold is

$$39 + 3 = 42.$$

Example: Geometric pigeonholes. Place 25 points in a 6 cm by 4 cm rectangle. Partition it into 24 unit squares, assigning boundary points consistently to one neighboring square. Two points share a square, so their distance is at most $\sqrt{1^2 + 1^2} = \sqrt{2} < 2$ cm.

Example: Queue threshold. A server distributes requests among 12 queues. At most $12 \cdot 8 = 96$ requests can avoid a queue of size nine. Thus

$$12(9 - 1) + 1 = 97$$

requests guarantee at least nine in one queue.

3 Worked examples

3.1 Cafe combo meals

A cafe offers five entrees, three vegetables, and five drinks. A meal contains one or two different entrees, one vegetable, and one drink. The order of the entrees does not matter.

Total meals. There are five single-entree choices and $5 \cdot 4/2 = 10$ pairs: each unordered pair is counted twice by selecting a first and second entree. Therefore,

$$(5 + 10) \cdot 3 \cdot 5 = 225 \text{ meals.}$$

Avoiding roast duck and coffee. Duck appears in five entree choices: alone or paired with one of the other four entrees. Subtract meals with duck and meals with coffee, then add back those containing both:

$$225 - \underbrace{5 \cdot 3 \cdot 5}_{75 \text{ with duck}} - \underbrace{15 \cdot 3}_{45 \text{ with coffee}} + \underbrace{5 \cdot 3}_{15 \text{ with both}} = 120.$$

Most frequently ordered meal. A customer orders one meal per day for 1000 days. Assign the 1000 orders to 225 meal types. Some type occurs at least

$$\left\lceil \frac{1000}{225} \right\rceil = 5$$

times. Consequently, the customer's most frequently ordered meal occurs at least five times. A specified meal, such as soy sauce chicken with cabbage and milk tea, could occur zero times unless it is known to be most frequent.

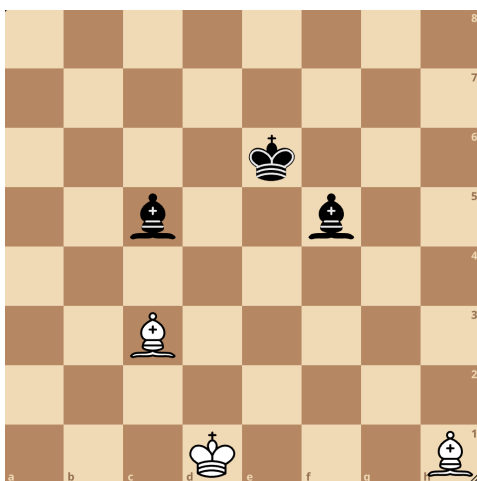
3.2 Registration codes

Codes have length one, two, or three. The first character is an uppercase letter; later characters may be uppercase letters or digits. Twelve otherwise valid codes are reserved. Count each length separately, add, and subtract:

$$\underbrace{26}_{\text{length 1}} + \underbrace{26 \cdot 36}_{\text{length 2}} + \underbrace{26 \cdot 36^2}_{\text{length 3}} - 12 = 34,646.$$

3.3 Two kings and four bishops

On an 8×8 chessboard, each player has one king, one bishop on light squares, and one bishop on dark squares. All six pieces occupy distinct squares. Ignore checks, king adjacency, and other chess rules initially.



One outcome assigns a square to each piece. The player and the bishop's square color distinguish the pieces, so no division is needed when they are placed in a fixed order. Choose an order, multiply the available choices, and split into disjoint cases only if necessary.

Bishops first. Place White's piece before Black's for each type:

Pieces	Choices
Two light-squared bishops	$32 \cdot 31$
Two dark-squared bishops	$32 \cdot 31$
Two kings	$60 \cdot 59$

After the bishops occupy four squares, exactly 60 remain. Thus

$$T = (32 \cdot 31)^2 \cdot 60 \cdot 59 = 3,483,586,560.$$

Kings first. This also works, but the available bishop squares depend on the kings' square colors:

White king	Black king	King placements	Light bishops	Dark bishops
light	light	$32 \cdot 31$	$30 \cdot 29$	$32 \cdot 31$
dark	dark	$32 \cdot 31$	$32 \cdot 31$	$30 \cdot 29$
light	dark	32^2	$31 \cdot 30$	$31 \cdot 30$
dark	light	32^2	$31 \cdot 30$	$31 \cdot 30$

Multiply across each row and add:

$$2(32 \cdot 31)^2(30 \cdot 29) + 2 \cdot 32^2(31 \cdot 30)^2 = 3,483,586,560.$$

Both orders give the same count. Bishops first avoids the four-case split.

Follow-up: Kings may not share an edge. Diagonal neighbors are allowed in this exercise; checks and other chess rules are still ignored. Subtract the forbidden placements from T .

For this count, place the kings first. Their squares form a horizontal or vertical domino. There are $8 \cdot 7 = 56$ locations of each orientation, and two orders for White's and Black's kings. Each pair occupies one light and one dark square, leaving 31 of each for the bishops. The forbidden count is

$$F = (56 + 56) \cdot 2 \cdot (31 \cdot 30)^2 = 193,737,600.$$

Hence the remaining number is

$$T - F = 3,289,848,960.$$

Here kings first is convenient because the restriction concerns their locations.

4 Practice

1. How many five-position decimal access codes are possible when every position may be any digit from 0 to 9?
2. A tour offers four walking routes at three start times or two boat routes at five start times. How many route-time plans are possible?
3. How many six-character uppercase-letter-or-digit passwords contain at least one digit?

4. How many unordered pairs of project partners can be chosen from 12 students?
5. List all complete sequences in a best-of-three playoff.
6. How many cards drawn without replacement guarantee (a) four of some suit and (b) four hearts?

Answers.

1. 10^5 .
2. $4 \cdot 3 + 2 \cdot 5 = 22$.
3. $36^6 - 26^6$.
4. $12 \cdot 11/2 = 66$.
5. $AA, ABA, ABB, BAA, BAB, BB$.
6. (a) $4(4 - 1) + 1 = 13$; (b) $39 + 4 = 43$.